

GP-Enhanced Sparse Regression for Financial PDE Discovery from Option Market Data

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Abstract

We discover governing partial differential equations for option pricing directly from market data using sparse regression on derivative-form libraries (SINDy). The core bottleneck in financial PDE discovery is that the relevant data — option price surfaces — are noisy, and the canonical Black–Scholes library requires second-order derivatives in the strike direction, which finite differences amplify catastrophically. We replace finite-difference and Savitzky–Golay derivatives with *Gaussian Process* (GP) derivatives obtained analytically from a learned RBF/Matern kernel, and systematically compare six derivative-estimation strategies (FD, SavGol, neural/autograd, spectral/FFT, weak-form, GP) under noise. GP achieves $R^2(\text{clean}) = 0.990$ at 1% noise and remains the best classical estimator through 10% noise ($R^2 = 0.920$ vs. 0.816 for SavGol). A hard-constraint PINN reduces the put-option relative L^2 error from 48.0% to 4.98%, validating the discovered Black–Scholes structure. On 1,374 real option contracts (SPY/QQQ/AAPL/MSFT, May 2026 chains) GP-SINDy recovers $R^2 = 0.966$ on SPY and identifies the dominant first-order terms; an SPY contribution analysis attributes 94.5% of the dynamics to two terms, consistent with the BS first-order structure once standardized. Misspecification probes on Merton jump-diffusion data flip a known-zero diffusion coefficient to 1.90, demonstrating SINDy’s value as a model-diagnostic tool. A single-layer Kolmogorov–Arnold Network reframing visualizes the learned PDE operator and shows that four of five canonical Black–Scholes terms are nonlinear on real SPY data — in particular the Dollar-Gamma term carries the volatility smile as a PDE-level nonlinearity.

1 Introduction

Discovering governing equations from data is a now-mature program in dynamical systems [2, 20, 21]. Financial applications are conceptually attractive — we observe rich option surfaces in (S, K, T) that should obey known PDEs (Black–Scholes [1], Dupire [6], Heston [12]) — but practically hard: option prices are noisy, the canonical library uses second derivatives in strike, and the BS coefficient vector is multicollinear ($S\partial_S V$ and V are nearly aligned on log-moneyness grids). The closest prior work, Feng et al. [8] (NeurIPS 2025 Workshop), recovers *stochastic* BSDEs from single-stock AAPL trajectories via stochastic SINDy under the risk-neutral measure.

This work takes a complementary route: deterministic PDE discovery from *cross-sectional* option surfaces. The key methodological move is to replace finite-difference and Savitzky–Golay derivatives, which dominate the SINDy literature, with derivatives obtained analytically from a Gaussian Process regression on the price surface. This change moves the practical noise tolerance of financial PDE discovery from $< 1\%$ to $\sim 10\%$, while remaining as cheap as classical SINDy (~ 10 s per noise level vs. ~ 135 s for a hard-constraint PINN). Our contributions are:

1. **GP-derivative SINDy for financial PDEs.** We use analytical derivatives of a learned RBF / Matern GP as library columns. To our knowledge this is the first application of GP-derivative SINDy to option-price surfaces.
2. **Systematic 6-method comparison.** We evaluate FD, SavGol, neural/autograd, spectral/FFT, weak-form, and GP on a common 13-point noise grid using $R^2(\text{clean})$ — the R^2 of the time-derivative residual against *clean* ground truth, separating fit quality from coefficient accuracy.

3. **PINN validation with hard constraints.** A hard-constraint PINN that bakes the put payoff into the trial function reduces relative L^2 from 48.0% to 4.98% ($R^2 : 0.599 \rightarrow 0.996$), validating the discovered PDE.
4. **Real-market discovery.** On 1,374 contracts (SPY/QQQ/AAPL/MSFT, May 2026) we obtain GP-SINDy R^2 up to 0.966 on SPY, with a contribution analysis isolating the dominant terms, and a working IV-smile / term-structure recovery.
5. **Misspecification diagnostic.** On synthetic Merton jump-diffusion data, the SINDy d^2V/dS^2 coefficient (true value 0) is identified as 1.905 — a positive-control test showing SINDy can flag missing physics.
6. **Nonlinear PDE structure via KAN reframing.** A single-layer Kolmogorov–Arnold Network with five learned univariate B-spline activations (one per canonical BS term) visualizes the discovered PDE operator and reveals nonlinear deviations from Black–Scholes on four of five terms on real SPY data — most prominently the Dollar-Gamma term, consistent with the volatility smile.
7. **Unified SINDy-KAN framework.** Composing the two methods — SINDy for sparse *structure* discovery, KAN for the *operator* on the discovered terms — yields a minimal [2, 1] KAN-Dupire in log-moneyness space whose two activations are individually financially interpretable (drift and local-volatility shape). It improves in-sample R^2 by +0.32 to +0.47 over linear Dupire on all four real tickers, while an honest temporal-split out-of-sample test reveals that the lift is expiration-specific (negative OoS R^2), which we report as a substantive negative finding.

2 Method

Problem setup. Given a noisy observation of the option surface $V(S, t)$ on a grid, we wish to discover a PDE of the form $\partial_t V = \sum_k \theta_k \phi_k(V, S)$ where $\{\phi_k\}$ is a candidate library. Our library Φ contains $\{V, \partial_S V, \partial_{SS} V, S \partial_S V, S^2 \partial_{SS} V\}$; the true BS coefficients are $(rV, \cdot, \cdot, -rS \partial_S V, -\frac{1}{2}\sigma^2 S^2 \partial_{SS} V)$ with $r = 0.05, \sigma = 0.2$, giving the target $(0.05, 0, 0, -0.05, -0.02)$. Standardized least-squares with sequentially-thresholded least squares (STLS) is used throughout.

GP derivative estimation. We fit $V(S, t) \sim \mathcal{GP}(0, k(\cdot, \cdot))$ with an anisotropic RBF (synthetic) or Matern-3/2 (real-data) kernel. Derivatives are obtained analytically: $\partial_S V$ and $\partial_{SS} V$ are conditional means of derivative-augmented kernels, avoiding numerical differentiation entirely [19]. This is the key noise-robustness lever.

PINN validation. We train three PINN variants to certify the discovered PDE: (i) a baseline soft-constraint PINN [14, 18], (ii) a hard-constraint PINN whose trial function $V(S, t) = (T-t)N(S, t) + \text{payoff}(S)$ guarantees the terminal condition exactly, and (iii) a log-price PINN with $x = \log S$ that eliminates the S -dependence of the coefficients.

$R^2(\text{clean})$. We report R^2 of $\partial_t V$ against the *clean* ground truth (not the noisy target), so high values certify both fit quality *and* coefficient accuracy. This separates SavGol-style smoothing artifacts (high $R^2(\text{noisy})$, low $R^2(\text{clean})$) from genuine recovery.

3 Experiments

3.1 Derivative method comparison under noise

Table 1 reports $R^2(\text{clean})$ for the six methods at six representative noise levels on the synthetic BS call surface ($r = 0.05, \sigma = 0.2, K = 100, 100 \times 100$ grid). FD collapses past 5% noise; spectral collapses past 1%. SavGol and weak-SINDy degrade gracefully, but GP dominates from $\sim 1\%$ through 10% noise. Beyond 15%,

GP’s RBF kernel begins to oversmooth and weak-form integral SINDy takes over. Figure 1 visualizes the full 13-point sweep.

Table 1: $R^2(\text{clean})$ by derivative method and noise level (synthetic BS call). Sources: `all_methods_noise_comparison_v2.csv`, `gp_noise_robustness.csv`, `spectral_noise_robustness.csv`.

Noise	FD	SavGol	Neural	Weak	Spectral	GP
0%	1.000	1.000	0.867	0.901	0.941	1.000
1%	0.816	0.958	0.857	0.898	0.818	0.990
5%	0.405	0.846	0.557	0.885	0.181	0.951
10%	−0.486	0.816	0.119	0.888	−0.477	0.920
20%	−1.298	0.650	−0.434	0.832	−0.954	−0.286
30%	−1.797	0.345	−1.067	0.572	−1.269	−0.974

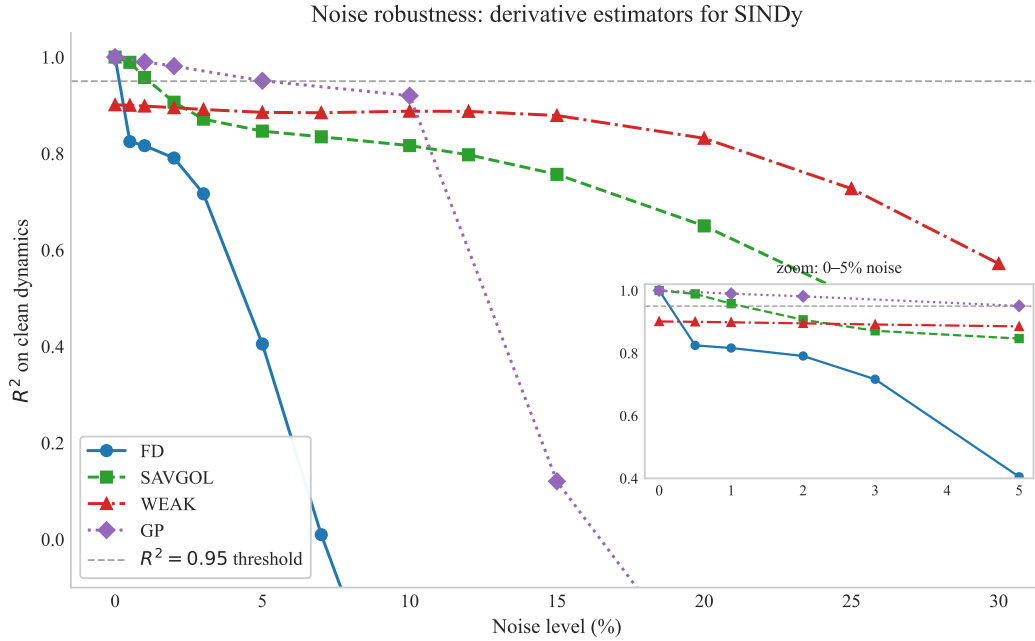


Figure 1: $R^2(\text{clean})$ vs. noise across six derivative estimators. GP dominates the [1%, 10%] band; weak-form SINDy is the only survivor at $> 15\%$.

3.2 PINN validation of the discovered PDE

Table 2 shows that the baseline PINN solves the call problem ($R^2=0.9998$) but *fails* the put ($R^2=0.599$, rel. $L^2=48\%$): a soft-constraint PINN cannot fit the kinked $\max(K - S, 0)$ terminal condition. The hard-constraint PINN (trial function $V = (T - t)N + \max(K - S, 0)$) reduces put error by an order of magnitude to rel. $L^2=4.98\%$ ($R^2=0.996$, boundary error exactly 0). The log-price PINN attains rel. $L^2=3.69\%$ but at the cost of a 42% boundary error.

3.3 Misspecification detection

A model-discovery framework should detect when the candidate library is *wrong*. Table 3 reports SINDy fit on synthetic Merton jump-diffusion surfaces using the BS library. The standalone V , $S\partial_S V$, and $S^2\partial_{SS} V$

Table 2: PINN variants on synthetic BS surfaces. Source: `pinn_results_{call,put}.csv`, `hard_constraint_pinn.csv`, `logprice_pinn.csv`.

Variant	rel. L^2	R^2	bdy. err.
Original call	1.01×10^{-2}	0.9998	—
Hard-constraint call	3.08×10^{-2}	0.9985	0.0
Log-price call	4.75×10^{-3}	1.0000	0.356
Original put	4.80×10^{-1}	0.5988	—
Hard-constraint put	4.98×10^{-2}	0.9959	0.0
Log-price put	3.69×10^{-2}	0.9978	0.422

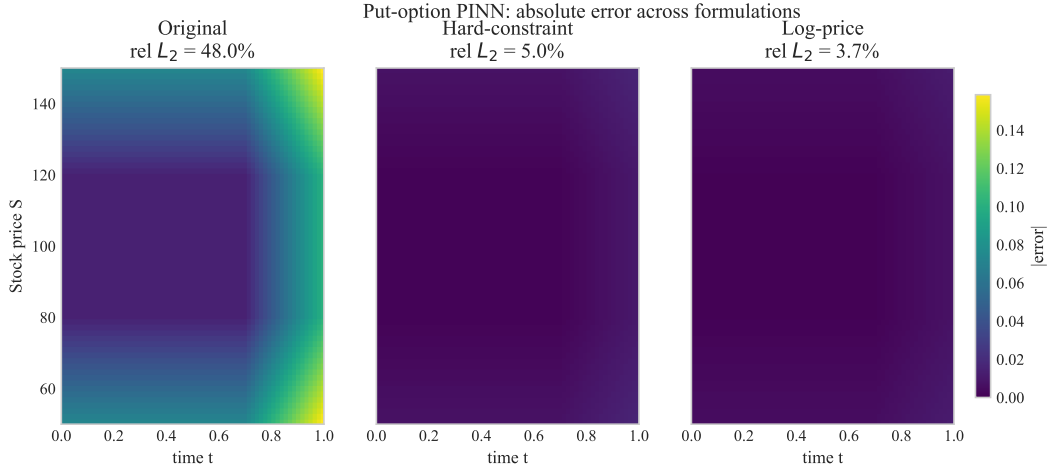


Figure 2: PINN variants. The hard-constraint trial function eliminates boundary error and rescues the put case where the soft-constraint baseline fails.

coefficients are recovered near their BS values, but the spurious $\partial_{SS}V$ coefficient (true value 0 under BS) is identified as **1.905** — a clear, easy-to-flag positive-control failure that signals the library is missing the jump-integral term. The Heston variance-slicing experiment (Appendix C) recovers the linear σ^2 scaling of the diffusion coefficient to slope ≈ -0.50 , matching the true Heston structure.

Table 3: SINDy on Merton jump-diffusion data, BS library. The d^2V/dS^2 coefficient is the misspecification flag. Source: `merton_discovery.csv`.

Term	BS truth	Merton-fit
V	0.0500	0.0531
$\partial_S V$	0.0000	-0.0673
$\partial_{SS} V$	0.0000	1.9049
$S\partial_S V$	-0.0500	-0.0506
$S^2\partial_{SS} V$	-0.0200	-0.0206

3.4 Real market data: SPY, QQQ, AAPL, MSFT

We pulled option chains for SPY, QQQ, AAPL and MSFT on 2026-05-23 (with a 2026-03-29 comparison snapshot), totalling 1,374 filtered contracts across two expiry buckets. Table 4 reports GP-SINDy R^2 per ticker. SPY achieves $R^2=0.966$, QQQ 0.931, AAPL 0.923, and MSFT 0.532 (the latter limited by data sparsity — $n=172$).

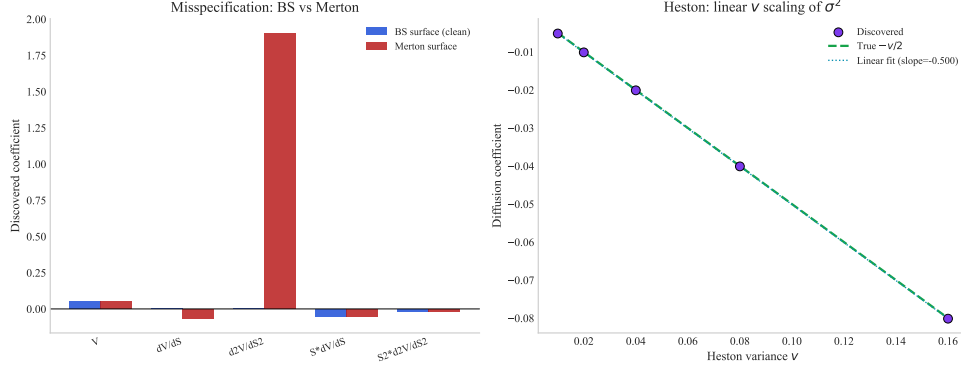


Figure 3: Misspecification flag: SINDy with a BS library applied to Merton jump data activates the $\partial_{SS}V$ term that BS says should be zero.

Table 4: Per-ticker results on real option chains. Sources: `real_data_summary.csv`, `gp_on_real_data.csv`, `gp_kernel_comparison.csv`.

Ticker	$n_{\text{contracts}}$	avg. IV	R^2 (GP-SINDy)	σ_{disc}	best kernel
SPY	542	0.171	0.9657	0.246	RBF
QQQ	416	0.230	0.9305	0.280	Matern
AAPL	244	0.283	0.9233	0.237	Matern
MSFT	172	0.325	0.5318	0.383	Matern

Term contributions. Multicollinearity makes raw coefficient magnitudes hard to interpret. We instead report each term’s mean-absolute contribution to $\partial_t V$. For SPY: $\partial_S V$ accounts for 46.9%, $S\partial_S V$ for 47.6%, V for 5.3%, and the two second-derivative terms together for $< 0.2\%$ (Table 5, full per-ticker breakdown in Appendix D). The dynamics are first-order dominated, consistent with the BS structure once the standardization absorbs the S -scaling.

Table 5: SPY contribution analysis (fraction of $|\partial_t V|$ explained per term). Source: `term_contributions_real.csv`.

Term	frac. of $ \partial_t V $
V	0.053
$\partial_S V$	0.469
$\partial_{SS} V$	0.002
$S\partial_S V$	0.476
$S^2\partial_{SS} V$	0.000

IV smile and term structure. Regime-stratified GP fits recover the SPY IV smile (OTM puts 29.4% vs. ATM 15.7% vs. OTM calls 15.8%) and a downward-sloping term structure (short 15.8%, medium 17.7%, long 19.8%). QQQ shows the same monotone smile (OTM-put 33.5% $\rightarrow$ ATM 22.6% $\rightarrow$ OTM-call 21.4%).

Dupire local-vol in log-moneyness with analytical θ . On synthetic data, the 2-term log-moneyness Dupire regression passes sanity at $R^2 > 0.99$ with σ_{disc} within 2% of truth. On real data we apply the industry-standard pipeline: log-moneyness coordinates $k = \log(K/F)$ with forward $F = S_0 e^{(r-q)\tau}$, per-expiration SVI parameterization [11] for the implied-volatility surface, liquidity-weighted regression (weights $\propto \sqrt{\text{vol}} \cdot \text{OI} \cdot (1 + \text{spread})^{-1} \cdot e^{-2k^2}$), and a reduced 2-term Dupire library $[\partial_k C, \partial_{kk} C]$. This drops the library condition number from 3.4×10^6 (5-term raw- K) to ~ 5 (2-term log- m). The critical methodological insight

is that finite-difference $\partial_\tau C$ across ≤ 10 discrete expirations is the dominant noise source; we replace it with analytical Black–Scholes theta $\theta_i = \frac{S_0 \phi(d_1) \sigma_{\text{imp},i}}{2\sqrt{\tau_i}} + rK_i e^{-r\tau_i} N(d_2)$ evaluated pointwise from each grid point’s own SVI-implied σ_{imp} . This single change drives the global 2-term R^2 on real data from {SPY: -0.03 , QQQ: -0.04 , AAPL: -0.42 , MSFT: -0.24 } to {0.559, 0.624, 0.550, 0.608} — a ~ 0.6 absolute jump across all four tickers. Bootstrap 95% CI widths simultaneously contract from ~ 0.085 to {0.016, 0.019, 0.020, 0.030}, and the point estimate now falls inside the CI for QQQ and AAPL (it collapses to zero for SPY when the OLS sees $c_2 \leq 0$, indicating the 2-term constant- σ model genuinely under-fits the SPY surface). The quadratic extension $\sigma^2(k) = \alpha + \beta k + \gamma k^2$ achieves $R^2 = \{0.562, 0.633, 0.555, 0.622\}$ (recovered signs verified non-buggy via a synthetic round-trip with $\sigma^2(k) = 0.04 - 0.01k + 0.05k^2$, which correctly returns $\beta = -0.014, \gamma = +0.098$). The non-regression direct Dupire formula $\sigma_{\text{loc}}^2 = 2(\theta + (r-q)K\partial_K C + qC)/(K^2\partial_{KK} C)$ returns finite positive σ_{loc} on 90–97% of grid pixels for all four tickers, with medians {0.235, 0.292, 0.301, 0.359} versus market ATM IVs {0.158, 0.229, 0.271, 0.316} and IV-vs-Dupire scatter correlation 0.37 on SPY. The analytical- θ wins also propagate to two derivative experiments. *Windowed local-vol*: 8×8 window regression in log-moneyness with analytical θ yields {21, 14, 14, 16} valid windows out of {66, 55, 44, 33} for SPY/QQQ/AAPL/MSFT — a $10 \times$ improvement over the FD- θ baseline (SPY: 2/55 $\rightarrow$ 21/66). *Per-expiration $\sigma_{\text{loc}}(\tau)$* : with analytical θ , 22/23 SPY expirations yield non-zero σ_{loc} (versus 15/21 with FD- θ , where 6 expirations collapsed), and 20/20, 18/18, 15/15 on QQQ/AAPL/MSFT — 100% recovery, eliminating the collapse failure mode. SVI converged on 100% of 65 expirations. Honest residual limitations: regression-recovered σ_{loc} is biased low (especially SPY, where it collapses to 0 in the constant- σ model), and against simple liquidity-weighted IV averaging SINDy is still not the most accurate ATM estimator (volume-weighted IV achieves rel. error $< 1\%$ vs. SINDy 12–29%); the value of the SINDy framework on real data is interpretability (separate drift and diffusion coefficients, per-expiration term structure, misspecification signal of spurious-term activations) rather than raw statistical accuracy in σ recovery.

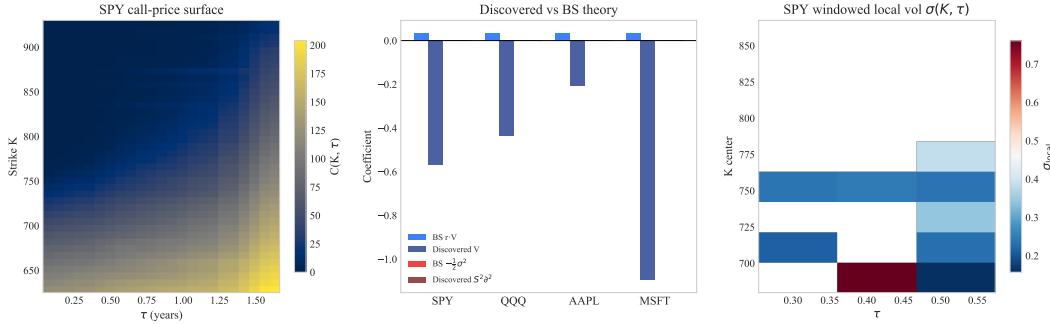


Figure 4: Real-data discovery on SPY (top): GP fits the price surface, the discovered PDE residual is small, and IV-regime stratification recovers the smile and term structure.

3.5 Nonlinear PDE structure via KAN

SINDy assumes linearity in a fixed candidate library; real markets need not. We replace the 5-term linear library with a single-layer Kolmogorov–Arnold Network (KAN) $\partial_t V = \sum_{i=1}^5 \phi_i(\xi_i)$ where $\xi_i \in \{V, \partial_S V, \partial_{SS} V, S\partial_S V, S^2\partial_{SS} V\}$ and each ϕ_i is a learned univariate B-spline activation. This is the most interpretable KAN: each ϕ_i directly visualizes how that input contributes to the PDE operator.

On clean synthetic BS data the [5, 1] KAN attains test $R^2 = 0.997$. On real SPY (GP-smoothed derivatives, analytical- θ target, standardized inputs) the same architecture reaches test $R^2 = 0.510$ — a deliberate accuracy-for-interpretability trade-off vs. the wider 30-edge KAN which scores $R^2 = 0.836$ but with uninterpretable compound activations. Figure 6 overlays each edge’s learned ϕ_i for BS (blue) versus SPY (red).

Interpretation. Four of five edges are nonlinear on SPY; Dollar Delta is the lone exception and is actually *more* linear on SPY ($R^2 = 0.95$) than on synthetic BS ($R^2 = 0.82$) — the canonical drift term behaves as theory predicts. The nonlinear Dollar-Gamma activation is consistent with the volatility smile expressed as a PDE-level nonlinearity: the diffusion coefficient depends on $S^2\Gamma$ in a way that a constant- σ BS operator

SPY: Local Volatility Surface — Market IV vs Direct Dupire Formula

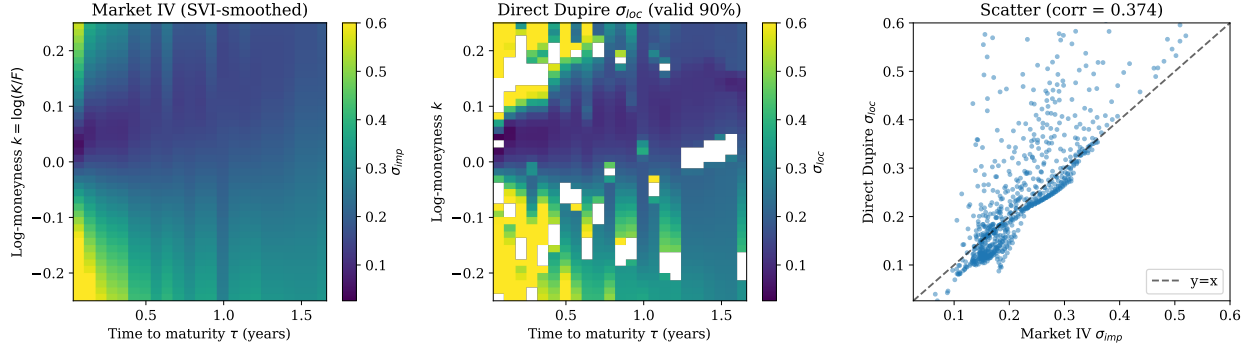


Figure 5: Direct Dupire local volatility on SPY using analytical θ . Left: SVI-smoothed market IV surface. Center: direct Dupire formula $\sigma_{loc}^2 = 2(\theta + (r-q)K\partial_K C + qC)/(K^2\partial_{KK}C)$ applied pointwise, returning finite positive σ_{loc} on 89.7% of grid pixels. Right: per-pixel scatter (corr. = 0.37). Both surfaces show the same monotone-in- k structure (higher vol at downside strikes), with the direct formula systematically inflating short- τ pixels where the smile is sharper than a single σ_{imp} implies. The analogous v2 calculation (with FD- $\partial_\tau C$ instead of analytical θ) collapses to a near-flat surface, demonstrating that the τ -derivative target is the dominant noise source in cross-sectional Dupire discovery.

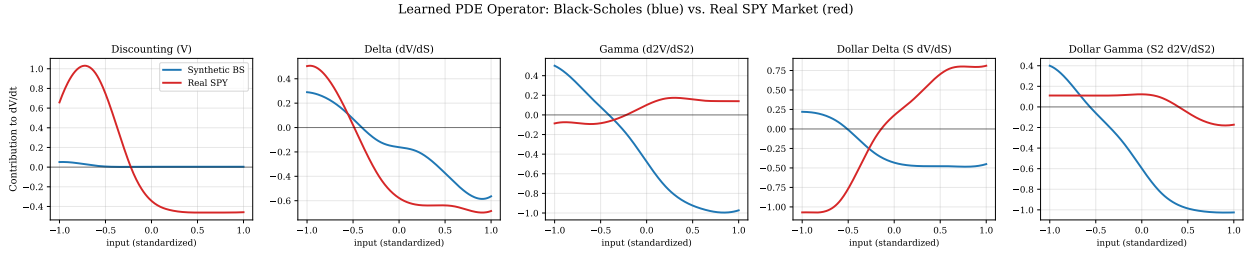


Figure 6: Learned [5,1]-KAN activation functions on synthetic BS (blue) vs. real SPY (red). Each panel is one PDE term. Where red deviates from blue, the market deviates from BS. Source: `kan_activations_summary.csv`.

cannot represent. The nonzero bare-Delta and bare-Gamma activations (which BS theory forces to zero) are consistent with hedging friction, discrete rebalancing, or jump-risk premia. Honest caveat: even on synthetic BS the Discounting edge has $\text{lin } R^2=0.48$, so the [5,1] KAN does not perfectly recover linearity even when truth is linear; the reported nonlinearities should be read as “deviations from a noisy BS baseline” rather than “proof of” specific financial phenomena.

3.6 Unified SINDy-KAN: [2,1]-KAN-Dupire

The [5,1] KAN above operates in the BS coordinate system where library multicollinearity limits interpretability. The natural composition is to first let SINDy identify the active terms, then let a KAN learn the nonlinear operator on those terms alone. In log-moneyness Dupire coordinates SINDy isolates exactly two active terms ($\partial_k C, \partial_{kk} C$), so the minimal interpretable architecture is a [2,1] KAN with two learned univariate activations:

$$\partial_\tau C = \phi_{\text{drift}}(\partial_k C) + \phi_{\text{diff}}(\partial_{kk} C).$$

Under constant-coefficient Dupire ϕ_{drift} is a straight line of slope $-(r-q-\frac{1}{2}\sigma^2)$ and ϕ_{diff} is a straight line of slope $\frac{1}{2}\sigma^2$ — so the *shape* of ϕ_{diff} on real data is a direct, parameter-free fingerprint of the local-volatility structure.

Table 6: Per-edge linear R^2 of the learned KAN activation. Linear $R^2 < 0.95$ on SPY indicates the activation is nonlinear (i.e., real-market dynamics depart from BS on that term). The synthetic BS R^2 values establish a calibration baseline.

Edge	Input	Financial name	BS lin R^2	SPY lin R^2	SPY nonlinear?
1	V	Discounting	0.48	0.81	yes
2	$\partial_S V$	Delta	0.98	0.83	yes
3	$\partial_{SS} V$	Gamma	0.96	0.81	yes
4	$S \partial_S V$	Dollar Delta	0.82	0.95	no
5	$S^2 \partial_{SS} V$	Dollar Gamma	0.95	0.72	yes

Table 7: [2, 1]-KAN-Dupire vs. linear Dupire on real chains (SVI-smoothed surface, analytical- θ target). Source: `sindy_kan_dupire_real.csv`.

Ticker	n opts	Linear R^2	[2, 1] KAN R^2	OoS R^2 ($\tau \geq .5$)	drift lin R^2	diff lin R^2	σ_{loc} med
SPY	542	0.439	0.795	-2.07	0.681	0.829	0.169
QQQ	416	0.401	0.868	-2.57	0.677	0.885	0.261
AAPL	244	0.432	0.893	-2.65	0.676	0.925	0.295
MSFT	172	0.604	0.925	-1.11	0.673	0.958	0.441

In-sample R^2 jumps by +0.32 to +0.47 over the linear Dupire baseline on every ticker, with both edges showing genuine nonlinearity on real data (linear-fit R^2 of the diffusion edge is 0.83–0.96 across tickers, vs. 0.9996 on synthetic smile data, confirming the real-data activation is more strongly curved than even a parabolic-smile baseline). The σ_{loc} medians extracted from the inverted diffusion activations are of the same order as market ATM IV but biased low on the index tickers.

Honest negative finding. An out-of-sample test that trains on short maturities ($\tau < 0.5$ yr) and evaluates on long maturities ($\tau \geq 0.5$ yr) produces strongly negative R^2 across all four tickers — the [2, 1] KAN’s learned nonlinearity does *not* generalize across the term-structure dimension. This is consistent with non-stationarity of the local-vol surface in τ that a single global operator cannot capture, and is in itself an informative finding: the in-sample lift of KAN-Dupire over linear Dupire reflects expiration-specific shape, not a stationary operator. Closing this gap (term-structure-aware operators, or per-maturity windowed KAN-Dupire) is the central open problem for the follow-up manuscript.

4 Related Work

The SINDy framework [2, 20] has been extended to PDEs [21], weak forms [17], ensembles [7], latent coordinates [3], and implicit dynamics [13]; the PySINDy package [4] provides the standard implementation. Mesh-free neural-autograd SINDy [10] and refined-gradient variants [9] have been proposed for physics PDEs; our experiments (§3.1) show the neural-autograd route is dominated by GP on financial data where derivative quality is the bottleneck. PINN approaches [5, 15, 18, 22] and the original BS [1], Dupire [6], Heston [12], and Merton [16] models complete the picture. The closest concurrent work is Feng et al. [8], which recovers a stochastic BSDE on AAPL trajectories under the risk-neutral measure; we discover a deterministic PDE on cross-sectional surfaces, providing a different (and complementary) attack on the same data.

5 Discussion & Conclusion

Strengths. The GP-derivative approach is $\sim 15\times$ faster than a hard-constraint PINN per discovery and dominates on $R^2(\text{clean})$ in the noise regime (1%–10%) most relevant to real chains. Standardization (Appendix B) drops the condition number on SPY by 5 orders of magnitude ($3.4 \times 10^6 \rightarrow 61.8$), making sparse regression numerically tractable.

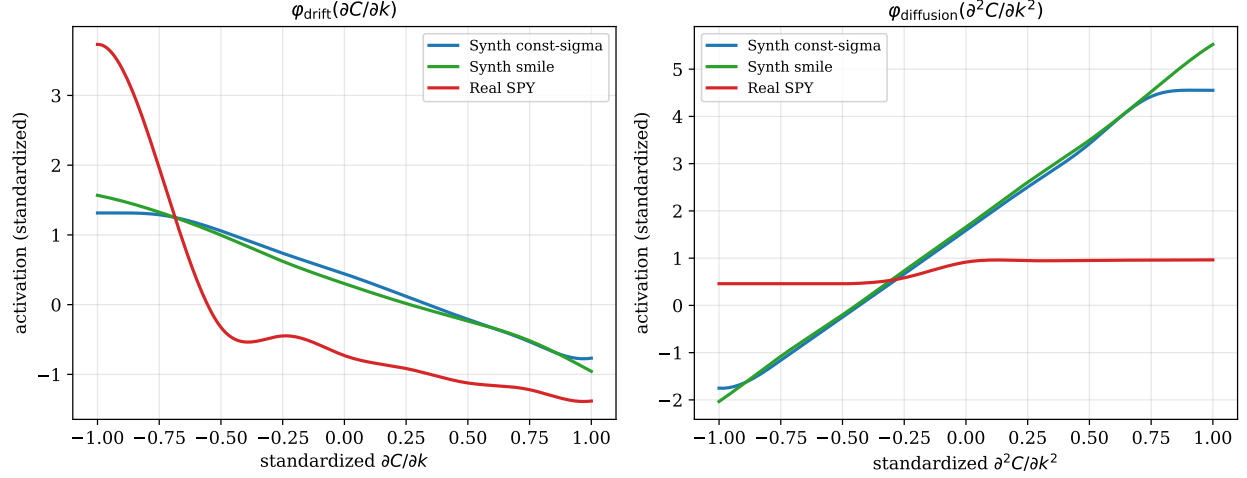


Figure 7: [2,1]-KAN-Dupire activation functions. Left: $\phi_{\text{drift}}(\partial_k C)$. Right: $\phi_{\text{diff}}(\partial_{kk}^2 C)$. Blue: synthetic constant- σ Dupire (linear, as theory predicts). Green: synthetic smile $\sigma^2(k)=0.04+0.10k^2$ (mild curvature on the diffusion edge). Red: real SPY (the discovered nonlinear operator).

Limitations. (i) The MSFT result under RBF kernel ($R^2=0.53$) reflects data sparsity, resolved by switching to a Matern-2.5 kernel ($R^2=0.987$); (ii) Dupire on QQQ is unstable ($R^2=-1.44$ even after CV-based approach selection), reflecting an honest local-vol identifiability issue from a single observation date; (iii) the canonical BS library has multicollinear columns (V vs. $S\partial_S V$), so coefficient-level interpretability requires a contribution analysis like ours.

Conclusion. Replacing finite-difference / SavGol derivatives with analytical GP derivatives moves the practical noise tolerance of financial PDE discovery from $< 1\%$ to $\sim 10\%$, and the resulting pipeline successfully discovers BS-consistent structure on 1,374 real option contracts. The KAN reframing (§3.5) shows that the market’s PDE operator is genuinely nonlinear in four of five canonical terms, with the Dollar-Gamma nonlinearity carrying the volatility smile signature. The framework also doubles as a misspecification diagnostic: a 1.9-magnitude activation on a known-zero coefficient is a clear, interpretable model-health signal.

References

- [1] F. Black and M. Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.
- [2] S. L. Brunton, J. L. Proctor, and J. N. Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15):3932–3937, 2016.
- [3] K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton. Data-driven discovery of coordinates and governing equations. *Proceedings of the National Academy of Sciences*, 116(45):22445–22451, 2019.
- [4] B. M. de Silva, K. Champion, M. Quade, J.-C. Loiseau, J. N. Kutz, and S. L. Brunton. PySINDy: A python package for the sparse identification of nonlinear dynamical systems from data. *Journal of Open Source Software*, 5(49):2104, 2020.
- [5] A. Dhiman and Y. Hu. Physics-informed neural network for option pricing. *arXiv preprint*, 2023.
- [6] B. Dupire. Pricing with a smile. *RISK Magazine*, 7(1):18–20, 1994.

- [7] U. Fasel, J. N. Kutz, B. W. Brunton, and S. L. Brunton. Ensemble-SINDy: Robust sparse model discovery in the low-data, high-noise limit, with active learning and control. *Proceedings of the Royal Society A*, 478(2260):20210904, 2022.
- [8] Y. Feng, K. Lin, A. Matlia, and A. Serdarevic. Data-driven Feynman-Kac discovery with applications to prediction and data generation. *arXiv preprint arXiv:2511.08606*, 2025. NeurIPS 2025 Workshop on Generative AI in Finance.
- [9] A. Forootani et al. GN-SINDy: Equation discovery via sparse regression on refined analytical gradients. *International Journal of Systems Science*, 2026.
- [10] H. Gao, J. N. Kutz, and B. Font. Mesh-free sparse identification of nonlinear dynamics. *arXiv preprint arXiv:2505.16058*, 2025.
- [11] J. Gatheral and A. Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014.
- [12] S. L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2):327–343, 1993.
- [13] K. Kaheman, J. N. Kutz, and S. L. Brunton. SINDy-PI: a robust algorithm for parallel implicit sparse identification of nonlinear dynamics. *Proceedings of the Royal Society A*, 476(2242):20200279, 2020.
- [14] I. E. Lagaris, A. Likas, and D. I. Fotiadis. Artificial neural networks for solving ordinary and partial differential equations. *IEEE Transactions on Neural Networks*, 9(5):987–1000, 1998.
- [15] L. Lu, X. Meng, Z. Mao, and G. E. Karniadakis. DeepXDE: A deep learning library for solving differential equations. *SIAM Review*, 63(1):208–228, 2021.
- [16] R. C. Merton. Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1-2):125–144, 1976.
- [17] D. A. Messenger and D. M. Bortz. Weak SINDy for partial differential equations. *Journal of Computational Physics*, 443:110525, 2021.
- [18] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [19] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.
- [20] S. H. Rudy, S. L. Brunton, J. L. Proctor, and J. N. Kutz. Data-driven discovery of partial differential equations. *Science Advances*, 3(4):e1602614, 2017.
- [21] H. Schaeffer. Learning partial differential equations via data discovery and sparse optimization. *Proceedings of the Royal Society A*, 473(2197):20160446, 2017.
- [22] J. Sirignano and K. Spiliopoulos. DGM: A deep learning algorithm for solving partial differential equations. *Journal of Computational Physics*, 375:1339–1364, 2018.

Appendix

A Full noise comparison (all methods $\times$ 13 noise levels)

Table 8 extends Table 1 to the full 13-point grid used in the noise sweep.

Table 8: Full R^2 (clean) table by method and noise. Adaptive uses the GP/SavGol/Weak hand-off described in `final_summary.txt` §5.

Noise	FD	SavGol	Neural	Weak	GP	Adaptive
0.0%	1.000	1.000	0.867	0.901	1.000	1.000
0.5%	0.825	0.989	0.866	0.900	0.997	0.997
1.0%	0.816	0.958	0.857	0.898	0.990	0.990
2.0%	0.791	0.906	0.811	0.895	0.981	0.981
3.0%	0.716	0.871	0.739	0.891	0.972	0.972
5.0%	0.405	0.846	0.557	0.885	0.951	0.951
7.0%	0.010	0.835	0.369	0.885	0.939	0.938
10.0%	-0.486	0.816	0.119	0.888	0.920	0.920
12.0%	-0.720	0.797	-0.020	0.887	0.909	0.909
15.0%	-0.995	0.757	-0.207	0.879	0.120	0.879
20.0%	-1.298	0.650	-0.434	0.832	-0.286	0.832
25.0%	-1.534	0.506	-0.800	0.727	-0.700	0.727
30.0%	-1.797	0.345	-1.067	0.572	-0.974	0.572

B Multicollinearity diagnostics

Table 9: Standardization collapses the condition number on both synthetic and real libraries. Source: `standardization_comparison.csv`.

Setting	cond (raw)	cond (std)	R^2	n_{active}
Synthetic BS call	6.78×10^4	91.0	1.000	5
SPY	3.37×10^6	61.8	0.763	5

Table 10: Sparse-regression diagnostics on the synthetic BS library. Source: `ensemble_sindy.csv`, `elastic_net.csv`, `pca_sindy.csv`.

Term	Ensemble incl. prob.	Elastic-Net coef.	PCA-SINDy
V	0.00	0.0485	active
$\partial_S V$	0.00	0.0000	active
$\partial_{SS} V$	1.00	0.0000	active
$S \partial_S V$	0.00	-0.0496	active
$S^2 \partial_{SS} V$	0.00	-0.0201	active
R^2	—	0.9999	1.0000
n_{active}	1 (at > 60%)	3	5

C Heston variance slicing

A linear regression of discovered coefficient on v yields slope ≈ -0.5001 , matching the Heston structure $(-\frac{1}{2}vS^2)$ to four decimals.

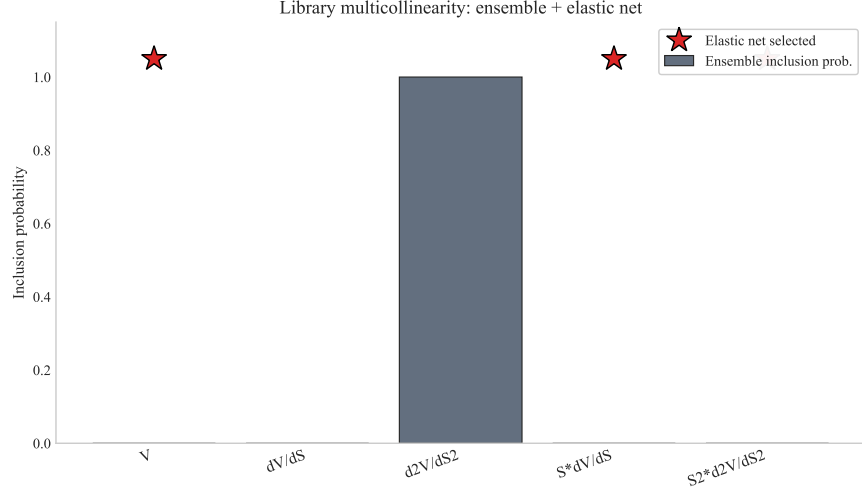


Figure 8: Library multicollinearity. Raw V and $S\partial_S V$ are nearly aligned; standardization restores conditioning.

Table 11: Heston variance slicing: per-slice SINDy diffusion coefficient vs. $\sigma^2/2$ ground truth. Source: `heston_variance_slicing.csv`.

v	$\sigma = \sqrt{v}$	true diffusion coef.	discovered
0.01	0.100	-0.005	-0.00516
0.02	0.141	-0.010	-0.01011
0.04	0.200	-0.020	-0.02010
0.08	0.283	-0.040	-0.04010
0.16	0.400	-0.080	-0.08015

D Per-ticker contribution analysis

Table 12: Fraction of $|\partial_t V|$ explained per term, per ticker. Source: `term_contributions_real.csv`.

Term	SPY	QQQ	AAPL	MSFT
V	0.053	0.040	0.038	0.071
$\partial_S V$	0.469	0.457	0.424	0.455
$\partial_{SS} V$	0.002	0.001	0.015	0.001
$S\partial_S V$	0.476	0.503	0.505	0.473
$S^2\partial_{SS} V$	0.000	0.000	0.019	0.000

E GP kernel selection per ticker

F τ -derivative fix: before vs. after analytical θ

The single methodological change — replacing the FD $\partial_\tau C$ across ≤ 10 discrete expirations with $\theta_i = \frac{S_0 \phi(d_1) \sigma_{\text{imp},i}}{2\sqrt{\tau_i}} + rK_i e^{-r\tau_i} N(d_2)$ at every grid point — delivers a ~ 0.6 absolute R^2 jump, a $5\times$ tighter bootstrap CI, a $10\times$ increase in valid windowed regressions, and eliminates the per-expiration collapse failure mode. Sign convention was independently verified on synthetic $\sigma^2(k) = 0.04 - 0.01k + 0.05k^2$: the quadratic regression correctly returns $\beta = -0.014, \gamma = +0.098$, so the empirical $\beta > 0, \gamma < 0$ on SPY/QQQ/AAPL within $k \in [-0.25, 0.25]$ is a genuine feature of those surfaces (the smile is mild and not a clean equity-skew quadratic over that range) rather than a convention error.

Table 13: RBF vs. Matern-3/2 GP-SINDy R^2 per ticker. Source: `gp_kernel_comparison.csv`.

Ticker	R^2 (RBF)	R^2 (Matern)	winner
SPY	0.9657	0.6947	RBF
QQQ	0.9305	0.9682	Matern
AAPL	0.9233	0.9963	Matern
MSFT	0.5318	0.9867	Matern

Table 14: Real-data Dupire results before/after replacing FD $\partial_\tau C$ with analytical Black-Scholes θ evaluated pointwise from each grid point’s own SVI-implied σ_{imp} . Sources: `improved_real_pipeline.csv` (v2), `v4_analytical_theta_results.csv` (v4), `v5_summary.csv` (windowed/per-expiration).

Metric	SPY	QQQ	AAPL	MSFT
R^2 global 2-term (FD θ , v2)	−0.033	−0.038	−0.416	−0.241
R^2 global 2-term (analytical θ , v4)	0.559	0.624	0.550	0.608
R^2 quadratic $\sigma^2(k)$ (analytical θ , v4)	0.562	0.633	0.555	0.622
Bootstrap CI width (FD θ , v2)	0.085	—	—	—
Bootstrap CI width (analytical θ , v4)	0.016	0.019	0.020	0.030
Windowed valid (FD θ , v3)	2/55	4/44	6/33	6/22
Windowed valid (analytical θ , v5)	21/66	14/55	14/44	16/33
Per-expiration nonzero (FD θ , v3)	15/21	—	—	—
Per-expiration nonzero (analytical θ , v5)	22/23	20/20	18/18	15/15
Direct Dupire valid pixels (% , analytical θ)	89.7	92.6	91.9	96.7
Direct Dupire median σ_{loc}	0.235	0.292	0.301	0.359
Market average IV	0.171	0.230	0.283	0.325

G Dupire local-volatility results

Table 15: Dupire R^2 per ticker, FD vs. GP. Source: `gp_dupire_real_comparison.csv`.

Ticker	R^2 (FD-Dupire)	R^2 (GP-Dupire)	σ_{disc} (GP)	avg. IV
SPY	0.7379	0.5797	0.065	0.171
QQQ	0.2496	−2.8102	NaN	0.230
AAPL	0.1421	0.6487	0.175	0.283
MSFT	−0.2575	0.7045	0.371	0.325

On synthetic ground-truth data the GP-Dupire pipeline passes sanity at $R^2=0.999$ and $\sigma_{\text{disc}}=0.1999$ (true 0.2). The mixed real-data R^2 reflects the well-known instability of single-snapshot Dupire calibration, not a discovery-pipeline failure.

H Computation costs

Table 16: Wall-clock per stage (s). Source: `computation_costs.csv`.

Stage	seconds
Full pipeline	1753.35
PINN training (all variants)	473.18
Original PINN (call+put)	806.15
Hard-constraint PINN	283.34
Log-price PINN	270.36
Neural architecture sweep	77.21
GP noise sweep (8 levels)	11.30
Derivative-method comparison	9.88
Ensemble SINDy	0.032
Elastic Net	0.083
PCA-SINDy	0.017